

CMAPF and Robot Soccer Simulation (Demo)

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Abstract

We demonstrate an integrated system combining Combinatorial Multi-Agent Path Finding (CMAPF) with a full-scale 11v11 robot soccer simulation. The CMAPF solver partitions target waypoints among robots and sequences them into open Hamiltonian tours, optimizing a linear combination of makespan and sum-of-costs. Seven solvers are evaluated on 20 benchmark maps with up to 8,000 agents; a free-cell memory optimization reduces RAM tenfold. The continuous variant uses Bézier-smoothed roadmaps with curvature-adaptive speed and analytical collision precomputation. The soccer simulation deploys learned PPO policies and a Kohonen-inspired strategy network under the full FIFA rule set, targeting deployment on Uniree G1 humanoid robots. Both components run on a single laptop core.

CMAPF Pipeline

The combinatorial variant of Multi-Agent Path Finding reduces to a multi-depot open Multiple Traveling Salesmen Problem (mTSP) on the waypoint distance graph. Given k robots with start positions $S = \{s_1, \dots, s_k\}$, n target waypoints $T = \{t_1, \dots, t_n\}$ with $n \gg k$, and a distance matrix d on SUT , the task is to partition T into k subsets and find an open Hamiltonian path for each robot minimizing $f = \alpha M + (1 - \alpha) \frac{1}{k} \sum_{r=1}^k C_r$, where M is makespan and C_r the cost of robot r 's tour.

Seven solvers are implemented and compared [1]:

- **SA**: Simulated annealing with swap and relocate neighborhoods, geometric cooling from $T_0 = 1000$ to $T_{\min} = 0.01$. With Voronoi initialization and 2-OPT refinement, it dominates the realistic Bézier pipeline (84.6% wins across 175 configurations on 11 maps).
- **k -medoids**: Balanced clustering with TSP open-path computation per cluster, balanced to within $\lceil n/k \rceil$ waypoints. Competitive on discrete grids; underperforms on weighted roadmaps.
- **NRPA**: Nested Rollout Policy Adaptation learns waypoint transition preferences through recursive Monte-Carlo rollouts. $I = 150000/n$ iterations per



Figure 1: CMAPF simulation on Boston_0.256: mid-execution with 4 robots visiting 100 waypoints.

level, $\ell_{\max} = 2$, temperature $\tau = 1/\ell$. Leads on constant-speed roadmaps (50% wins).

- **LKH-3**: Lin–Kernighan–Helsgaun heuristic in both clustered (1-TSP per robot) and full integrated (multi-depot open mTSP) modes.
- **Branch-and-Bound**: Exact solver for $n \leq 22$, with Held–Karp dynamic programming lower bounds for pruning. Matches SA optimum on small instances.
- **AMAPF / DMAPF**: BFS-based MAPF on the mTSP assignment, swapping precomputed shortest-path trees on collision. DMAPF adds dynamic replanning when agents finish tour segments early.

Figure 1 shows a typical CMAPF execution. Under the mild assumption that only opposing edge and node conflicts occur on the overlay of assigned shortest paths, CMAPF and AMAPF are optimal with respect to the underlying waypoint-to-robot assignment.

In the continuous variant [1], the grid is replaced by a roadmap graph generated by adaptive distance-transform sampling from Moving AI benchmark maps. Robot paths are smoothed via cubic Bézier curves with C^1 continuity and arc-length reparameterization.



Figure 2: Simulation dashboard (2026 update) showing 11v11 4-3-3 gameplay with real-time Q-magnitude metrics, team distance tracking, tournament group standings, knockout bracket, and actor-critic policy labels.

Speed is curvature-adaptive: $v(s) \leq \sqrt{a_{\max}/\kappa(s)}$, bounding centripetal acceleration. Collisions are resolved by *identity exchange* at analytically precomputed meeting points on graph edges, proven to be the exact elastic solution for equal-mass point particles. Dijkstra’s algorithm replaces BFS on the weighted roadmap, guaranteeing optimal CMAPF solutions when the mTSP is solved exactly. Delta encoding and binary frame compression achieve 330× and 19× reductions, keeping the full benchmark suite under 1 GB.

Robot Soccer Simulation

The soccer simulation [2] runs at 20 Hz on a 105×68 m regulation pitch enforcing the full FIFA rule set: off-side, fouls, cards, corners, throw-ins, goal kicks, and penalties. Ball physics includes gravity ($g = 9.81 \text{ m/s}^2$), Magnus effect, air drag, and rolling friction. Players accelerate at 3.5 m/s^2 to 3.0 m/s maximum, with PES-style skill attributes generated with role-appropriate bias. All outcomes—shots, passes, saves, tackles—are determined by simulation physics rather than probabilistic awards, giving RL policies continuous reward gradients.

Strategy Network. A Kohonen-inspired network deforms a 4-4-2 formation in response to ball position $\mathbf{b}$ and possession flag π . For each formation node i with base position $\mathbf{p}_i$, a weight $w_i = \text{clamp}(1 - \|\mathbf{p}_i - \mathbf{b}\|/D_i, w_i^{\min}, w_i^{\max})$ and attraction point $\mathbf{r}_i$ are computed, yielding $\mathbf{p}'_i = \mathbf{p}_i + w_i(\mathbf{r}_i - \mathbf{p}_i)$. Width modulation stretches or compresses formation edges laterally based on ball position; depth compression adjusts vertical spacing between lines, pressing high in possession and dropping back when defending. The network produces target positions for all 11 players in a single forward pass.

PPO Policy. The dashboard in Fig. 2 visualizes the simulation with tournament standings and

Q-magnitude metrics. A 3-headed actor-critic policy (768 parameters, 32-D observation per player, 64-unit ReLU trunk) selects movement targets and action logits (shoot, pass, dribble, move), executing in under a millisecond on embedded CPUs. Training uses PPO-Clip with GAE advantage estimation; the value head trains via TD(0) on every timestep for dense learning signal. A hand-crafted tactical prior (counter-attack, gegen-pressing, overload-to-switch) is blended during training with weight annealing from 1.0 to 0.3. Neural policies substantially outperform tabular Q-learning for spatial positioning, driven by continuous generalization of pairwise-distance features.

Hardware Deployment. The PPO policy is the upper layer of a two-level stack targeting the Unitree G1 humanoid: it outputs world-frame (x, y) targets to a SONIC motor-control PPO running at 200Hz. Both layers share the clipped-surrogate algorithm, with feasibility feedback flowing upward. Curriculum transfer (2v2 $\rightarrow$ 3v3 $\rightarrow$ 5v5 $\rightarrow$ 11v11) progressively increases complexity.

Demonstration

The demo shows both systems live on a single laptop. An operator launches a CMAPF benchmark run on a selected map (e.g., Boston_0_256 with 100 agents), observing real-time visualization with Bézier-smoothed trajectories, curvature-adaptive speed profiles, and collision resolution. Switching to the soccer simulator, the operator watches an 11v11 match with the learned PPO policy against a scripted opponent, with the strategy network’s formation deformation visible in the browser dashboard. Curriculum transfer from 3v3 through 5v5 to 11v11 can be demonstrated step by step, showing how the neural policy transfers positional features across team sizes.

Full technical descriptions appear in [1] and [2]. Parts of the work will also appear in [3].

References

- [1] S. Edelkamp. Combinatorial and dynamic multi-agent path finding: Exact and heuristic solvers with scalable visualization. Unpublished manuscript. Available from author, 2026.
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